Lecture 19 Practice Questions and Solutions

Question 1 (q1_sol.py)

A small campus cafe stores each customer's order in two lists, e.g.:

```
items  = ['latte', 'muffin', 'tea']
prices = [5.25, 3.5, 2.75]
```

items[i] is the name of something the customer ordered, and prices[i] is its price. So the latte costs 5.25, the muffin 3.50, and the tea $2.75.

Write a function print_receipt(items, prices) that prints a receipt to the screen:

- The first line is the cafe name: The Java Hut Cafe
- The following lines have the form "item: price", i.e. each item and its price.
- The last line is the total of all the prices: "Total: total"

Show all prices with exactly 2 digits after the decimal point. The function print to the screen and not return anything.

Sample Run

If items and prices are as above, calling print_receipt(items, prices) should print:

```
Java Hut Cafe

latte: $5.25
muffin: $3.50
tea: $2.75

Total: $11.50
```

Sample Solution

```
def print_receipt(items, prices):
    print("The Java Hut Cafe\n")
    num_items = len(items)
    for i in range(num_items):
        item = items[i]
        price = prices[i]
        print(f"{item}: ${price:.2f}")
    total = sum(prices)
    print(f"\nTotal: ${total:.2f}")

#
# main program
```

```
#
items  = ['latte', 'muffin', 'tea']
prices = [5.25, 3.5, 2.75]
print_receipt(items, prices)
```

Question 2 (q2_sol.py)

Modify the previous program so that in addition to printing the receipt on the screen, it also writes the receipt to a text file called "receipt.txt". The contents of "receipt.txt" should be the same as the printed output.

Sample Run

Suppose items and prices are defined like this:

```
items = ['latte', 'muffin', 'tea']
prices = [5.25, 3.5, 2.75]
```

Then calling print_receipt(items, prices) prints the following both on the screen, and to the file "receipt.txt":

```
Java Hut Cafe

latte: $5.25
muffin: $3.50
tea: $2.75

Total: $11.50
```

Sample Solution

```
def print_receipt(items, prices):
    output_file = open("receipt.txt", "w")  # "w" means open for writing
    print("The Java Hut Cafe\n")
    output_file.write("The Java Hut Cafe\n\n") # note the extra \n at the end
    num_items = len(items)
    for i in range(num_items):
        item = items[i]
        price = prices[i]
        print(f"{item}: ${price:.2f}")
        output_file.write(f"{item}: ${price:.2f}\n") # note the \n at the end
    total = sum(prices)
    print(f"\nTotal: ${total:.2f}")
    output_file.write(f"\nTotal: ${total:.2f}\n") # note the \n at the end
    output_file.close()
#
# main program
#
items  = ['latte', 'muffin', 'tea']
prices = [5.25, 3.5, 2.75]
print_receipt(items, prices)
```

Question 3 (q3_sol.py)

Weekly Step Tracker

Your fitness tracker shows how many steps you took each day, and you want a quick summary of your week.

Write a program that asks the user for their step count on each of 7 days. Use input() and append each number to a list called steps. Then it prints:

- the total number of steps for the week
- the fewest steps taken on any one day and the most steps taken on any one day
- the average steps per day, rounded to 1 digit after the decimal point
- all 7 step counts from smallest to largest, one per line

Sample Run

```
Steps on day 1: 6421
Steps on day 2: 10250
Steps on day 3: 3890
Steps on day 4: 8125
Steps on day 5: 12034
Steps on day 6: 7560
Steps on day 7: 5013

 Total steps: 53293
Fewest steps: 3890
  Most steps: 12034
     Average: 7613.3

Smallest to largest:
3890
5013
6421
7560
8125
10250
12034
```

Sample Solution

```
#
# read in the step counts from the user
#
steps = []
for i in range(7):
    n = int(input(f'Steps on day {i + 1}: '))
    steps.append(n)
```

```
#
# calculate and print the stats
#
total = sum(steps)
least = min(steps)
most = max(steps)
average = total / len(steps)
print()
print(f' Total steps: {total}')
print(f'Fewest steps: {least}')
print(f'  Most steps: {most}')
print(f'     Average: {average:.1f}')

#
# print the step counts from smallest to largest
#
steps.sort()
print()
print('Smallest to largest:')
for n in steps:
    print(n)
```